



Name: _____

Class: _____

Date: _____

Mr. Rawson • IB DP Physics

Simple Harmonic Motion 4 – Analysing Your Oscillation Data

Linearizing, and Position–Velocity–Acceleration

You already have the numbers. This lesson turns your spring–mass results into a spring constant, checks your prediction against your own measurement, and uses your own oscillator to work through how displacement, velocity and acceleration relate. Keep your practical packet and spreadsheet open — every question below uses *your* data, not a supplied one.

Linearizing: from your graph to a spring constant

Key concept

$T = 2\pi\sqrt{m/k}$ is a square-root relationship — a graph of T against m curves, and a curve's gradient changes from point to point, which makes it useless for finding k . Squaring both sides straightens it:

$$T^2 = \frac{4\pi^2}{k} m$$

T^2 against m is now a straight line through the origin (plus a small manufacturing offset, more below), with gradient $4\pi^2/k$. This is **linearizing**: choosing the version of a relationship whose graph is straight, because a straight line is the one shape whose gradient is a single, measurable number.

- 1) **State** why plotting T^2 against m lets you find k , when plotting T against m does not.

[2]

Worked example — finding k from a gradient

A group's T^2 vs m graph has gradient $0.51 \text{ s}^2/\text{kg}$.

$$\text{gradient} = \frac{4\pi^2}{k} \Rightarrow k = \frac{4\pi^2}{\text{gradient}} = \frac{4\pi^2}{0.51} = 77.4 \text{ N/m}$$

Check: a stiffer spring gives a smaller gradient. If your k comes out smaller than a spring you can barely stretch by hand should have, re-read your gradient before going further.

2) **Calculate** k from the gradient of your own T^2 vs m graph. Show the rearrangement.

[3]

My gradient = _____ s^2/kg

k = _____ N/m

3) **Explain** why your graph has a small positive T^2 -intercept instead of passing through the origin, and state whether it affects the value of k you just calculated.

[3]

4) **Compare** your calculated k against a direct measurement: pick one of your six masses, use your k to predict its period, then compare that prediction with the period you actually measured for that mass.

[4]

Mass chosen, m = _____ kg

Measured T = _____ s

Predicted $T = 2\pi\sqrt{m/k} =$ _____ s

Percentage difference between predicted and measured: _____ %

What the amplitude check told you

5) **State** the pattern in your three amplitude-check readings — the same mass, three different pull-down distances.

[2]

6) **Connect** your amplitude-check result to the formula $T = 2\pi\sqrt{m/k}$. What feature of the formula does your result confirm?

[2]

Position, velocity and acceleration — for your own oscillator

Key concept

Three quantities describe an oscillator at any instant: displacement x , velocity v , and acceleration $a = -\omega^2 x$. Knowing one at a given moment, together with ω , gets you the others. You already have everything needed to find ω for your own spring.

7) **Calculate** ω for your oscillator, using the period you measured for the mass you chose in question 4.

[2]

$$\omega = \frac{2\pi}{T} = \text{_____ rad/s}$$

(a) You pulled the mass down by roughly 2 cm before releasing it — that was your amplitude, x_0 . Using your own ω , calculate the acceleration at the instant of release, when $x = x_0$.

[3]

(b) State the acceleration at the instant the mass passes through the equilibrium position, and explain why.

[2]

8) **Complete** the table below for your own oscillator, at the three points in one full cycle listed.

[6]

Point in cycle	Displacement x	Velocity v	Acceleration a
Released, $x = x_0$			
Passing equilibrium			
Other extreme, $x = -x_0$			

Higher level only

Speed at a chosen displacement.

$$v = \pm\omega\sqrt{x_0^2 - x^2}$$

9) **Calculate** the speed of your own oscillator at the instant its displacement is half its amplitude, $x = x_0/2$. Use your own ω and $x_0 = 0.020$ m.

[4]